

ECON 3350/7350 Applied Econometrics for Macroeconomics and Finance

Lecture 11: Cointegration - The Multiple Equation Case

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Recommended Readings

AUTHOR	TITLE	CHAPTER	CALL No
Enders, W	Applied Econometric Time Series, 4e	6.3 to end of chapter	HB139 .E55 2015
Verbeek, M	A Guide to Modern Economet- rics, 4e	9.5	HB139 .V465 2012

VARs

- ▶ The SVAR

$$Bx_t = \Gamma_0 + \Gamma(L)x_{t-1} + \epsilon_t$$

- ▶ The VAR is given by

$$x_t = A_0 + A(L)x_{t-1} + e_t \quad (1)$$

- ▶ where $A(L) = A_1 + \dots + A_p L^{p-1}$

$$[I_n - A(1)]x_t = A_0 + e_t \quad (2)$$

- ▶ If $[I_n - A(1)] = I_n - A_1 - \dots - A_p$ is **not invertible**
 - ▶ It **cannot be** the case that **all variables** in x_t are **stationary** $I(0)$ series.
 - ▶ At least one stochastic trend must be present.

Cointegration

Definition

The components of the vector x_t are said to be cointegrated of order d , b , denoted by $x_t \sim CI(d, b)$ if

1. All components of x_t are integrated of order d ($d > 0$)
 2. There exists a vector β such that the linear combination $\beta'x_t = z_t$ is integrated of order $(d - b)$, $d - b \geq 0$
- ▶ Normally each element of x_t is $I(1)$, and the linear combinations are $I(1 - 1)$, that is $I(0)$ or stationary.
 - ▶ When the **cointegrating rank**, $r > 1$, β' is a **cointegrating matrix** of dimension $r \times n$

Cointegration (cont.)

- ▶ Some important issues regarding this definition:
 - ▶ Cointegration refers to a **linear** combination of non-stationary variables.
 - ▶ A cointegrating vector is **not unique**
 $\beta'x_t = z_t$ or $(\kappa\beta)'x_t = \kappa z_t$, $\kappa > 0$, are both cointegrating vectors.
 - ▶ Testing will allow us to **identify** r , the **cointegrating space**.
 - ▶ The **vectors** are only **identified up to a normalization**.

Example

$n=3$, Two cointegrating vectors, $r=2$

$$x_t = \begin{pmatrix} x_{1t} \\ x_{2t} \\ x_{3t} \end{pmatrix}$$

$$z_t = \begin{pmatrix} 1 & \beta_{21} & \beta_{31} \\ \beta_{12} & 1 & \beta_{32} \end{pmatrix} \begin{pmatrix} x_{1t} \\ x_{2t} \\ x_{3t} \end{pmatrix} = \begin{pmatrix} x_{1t} + \beta_{21}x_{2t} + \beta_{31}x_{3t} \\ \beta_{12}x_{1t} + x_{2t} + \beta_{32}x_{3t} \end{pmatrix} \sim I(0)$$

Vector Error Correction Model

Example

$n=3, p=2$, Two cointegrating vectors, $r=2$

$$z_t = \begin{pmatrix} 1 & \beta_{21} & \beta_{31} \\ \beta_{12} & 1 & \beta_{32} \end{pmatrix} \begin{pmatrix} x_{1t} \\ x_{2t} \\ x_{3t} \end{pmatrix} = \begin{pmatrix} x_{1t} + \beta_{21}x_{2t} + \beta_{31}x_{3t} \\ \beta_{12}x_{1t} + x_{2t} + \beta_{32}x_{3t} \end{pmatrix} \sim I(0)$$

Then,

$$x_t = A_0 + A_1x_{t-1} + A_2x_{t-2} + e_t \quad (3)$$

$$\Delta x_t = \pi_0 + \pi x_{t-1} + \pi_1 \Delta x_{t-1} + e_t$$

$$\pi = -(I_n - A_1 - A_2) = \alpha\beta'$$

$$\pi_1 = -A_2$$

$$\Delta x_t = \pi_0 + \alpha z_{t-1} + \pi_1 \Delta x_{t-1} + e_t$$

- ▶ The Δx_t and e_t are stationary (by assumption), then it must be the case that πx_t is also stationary.
- ▶ This could reflect **three different situations**
 - ▶ If **all elements in x_t are stationary $I(0)$ variables**, the matrix $\pi = -[I_n - A(1)]$ must be of full rank and invertible so that we can write a VMA representation, **and** (1) is a stationary **VAR model for x_t**
 - ▶ If **all elements in x_t are integrated** of order one and no cointegrating relationships exist, it must be the case the $\pi = 0$ and (1) is a stationary **VAR model for Δx_t** .
 - ▶ If **π is of rank r ($0 < r < n$)**, the elements in πx_t are **linear combinations that are stationary**. If the variables in x_t are $I(1)$, these **linear combinations** must correspond to **cointegrating vectors**. π can be written as a product of a $n \times r$ matrix α and an $r \times n$ β' that both have rank r ; and (1) is a **VEC model for Δx_t** .

VECM (cont.)

- ▶ When $\pi = \alpha\beta'$ the model becomes

$$\Delta x_t = \pi_0 + \alpha\beta'x_{t-1} + \pi_1\Delta x_{t-1} + \dots + \pi_{p-1}\Delta x_{t-p+1} + e_t \quad (4)$$

$$= \pi_0 + \alpha z_{t-1} + \pi_1\Delta x_{t-1} + \dots + \pi_{p-1}\Delta x_{t-p+1} + e_t \quad (5)$$

- ▶ which is known as a Vector Error Correction Model (VEC)
- ▶ The linear combinations $z_t = \beta'x_t$ present the r cointegrating relationships, known as the **cointegrating rank**.
- ▶ If $r < n$ then it must be the case that some of the n variables in x_t have **common stochastic trends**.
- ▶ The coefficients in **α measure** how the elements in Δx_t are **adjusted** to the r "equilibrium errors."

Residual Based Tests

- ▶ Engle-Granger Approach
- ▶ Simple to implement. Based on a regression.
 - ▶ Can be used to indicate whether cointegration is present
 - ▶ Cannot indicate the cointegration rank (it will estimate a linear combination of the r cointegrating vectors)
 - ▶ Can be sensitive to the choice of variable used on the left-hand side.
- ▶ Covered in [Single Eq. Cointegration](#)

Residual Based Tests (cont.)

Engle-Granger Test. Under the Null the residuals of the cointegrating regression are integrated (*No Cointegration*)

- ▶ **Tests Hypothesis:** H_0 : *No cointegration*
- ▶ **Computed Test Statistic:**
 1. Regress x_{1t} (first variable in x_t) on the remaining $n - 1$ variables (cointegrating regression). Choice of x_{1t} does not matter in theory.
 2. Test the residuals of the cointegrating regression for a unit root
- ▶ **Critical Values:**
 - ▶ use Engle-Yoo critical values- Table C (Enders), p-values provided by EViews 9.
- ▶ **Decision:**
 - ▶ Reject H_0 : No Cointegration.
 - ▶ Conclude there exist at least one cointegrating vector amongst the n variables in the system.

VECM Testing Approaches

- ▶ Johansen Tests (based on VARs)

- ▶ Maximum likelihood approach to estimate the VEC model and test for the cointegration rank.
- ▶ Two test statistics:
 - ▶ Trace and Maximum Eigenvalue.
 - ▶ They can give different results.
- ▶ Sensitive to the specification of deterministic components
- ▶ Sensitive to the choice of lag length of the VAR

Johansen Tests for cointegration

- ▶ Based on estimating a VEC by maximum likelihood

$$\Delta x_t = \pi_0 + \pi x_{t-1} + \pi_1 \Delta x_{t-1} + \dots + \pi_{p-1} \Delta x_{t-p+1} + e_t \quad (6)$$

where $e_t \sim NID(0, \Sigma)$ as we need to assume a particular distribution.

- ▶ The number of cointegrating vectors is equal to the rank of π , denoted r .
- ▶ For $(0 < r < n)$, we can write $\pi = \alpha\beta'$ where α and β are $n \times r$.
- ▶ Johansen's approach is to estimate (6) for a given value of r .
- ▶ The rank of a matrix is equal to the number of its non-zero characteristic roots. This means we can test hypotheses concerning r using the estimated characteristic roots of π .

Johansen Tests for cointegration (cont.)

► Trace Test Statistic

- We reject $H_0 : r = r^*$ in favor of $H_1 : r > r^*$ if the value of a trace test statistic is greater than critical values found in Johansen-Juselius tables.

► Maximum Eigenvalue (Lambda Max) Test Statistic

- We reject $H_0 : r = r^*$ against $H_1 : r = r^* + 1$ if the value of a maximum eigenvalue test statistic is greater than critical values in Johansen-Juselius tables.
- Both tests have a non-standard chi-squared distribution
- They are sequential tests

Johansen Tests for cointegration (cont.)

- ▶ Important **considerations** when testing for cointegration and estimating a VEC **using Johansen's estimation approach**
 - ▶ Underfitting the model distorts this test severely. Use AIC, SC, LR to **determine the lag length “ p ”, using the VAR in levels not the VECM**
 - ▶ Intercepts in the VEC and/or inside the cointegrating relationship can change the outcome of the cointegration test as the distribution of the test statistic depends on nuisance parameters similarly to the the Dickey-Fuller test.

VEC and Deterministic Components

- ▶ Nuisance parameters lead to non-standard distributions for the cointegration tests
- ▶ Alternative specifications include:
 - ▶ No constant and/or trend

$$\Delta x_t = \alpha \beta' x_{t-1} + \pi_1 \Delta x_{t-1} + \dots + \pi_{p-1} \Delta x_{t-p+1} + e_t$$

- ▶ Unrestricted constant and/or trend in the VEC

$$\Delta x_t = \pi_{10} + \delta_1 t + \alpha \beta' x_{t-1} + \pi_1 \Delta x_{t-1} + \dots + \pi_{p-1} \Delta x_{t-p+1} + e_t$$

- ▶ Restricted constant and/or trend in the cointegrating vector

$$\Delta x_t = \alpha (\beta_{10} + \delta_2 t + \beta' x_{t-1}) + \pi_1 \Delta x_{t-1} + \dots + \pi_{p-1} \Delta x_{t-p+1} + e_t$$

- ▶ Unrestricted and Restricted constant and/or trend

$$\begin{aligned} \Delta x_t = & \pi_{20} + \delta_1 t + \alpha (\beta_{20} + \delta_2 t + \beta' x_{t-1}) + \pi_1 \Delta x_{t-1} + \dots \\ & + \pi_{p-1} \Delta x_{t-p+1} + e_t \end{aligned}$$

Example

Testing all-lives govt bonds clean price indexes

AAUGVAL (A\$) : AUSTRALIAN

AJPGVAL (Y) : JAPAN

ABDGVAL (E) : GERMANY

AUKGVAL (£) : UNITED KINGDOM

AUSGVAL (US\$) : UNITED STATES

- ▶ The All-bond index covers all traded bonds, irrespective of liquidity, and is primarily of interest to long term investors.
- ▶ The price excluding accrued interest is known as the "flat" or "clean price".
- ▶ Data sourced from Datastream

Determining the Lag-Length

Lag Length - Test

VAR Lag Order Selection Criteria

Endogenous variables: AAUGVAL ABDGVAL AJPGVAL AUKGVAL AUSGVAL

Exogenous variables: C

Sample: 1986M09 2006M09

Included observations: 224

Lag	LogL	LR	FPE	AIC	SC	HQ
0	-3210.775	NA	2028674.	28.71228	28.78843	28.74302
1	-1694.433	2951.452	3.344783	15.39672	15.85364*	15.58116*
2	-1662.685	60.37761	3.150258*	15.33647*	16.17416	15.67460
3	-1645.114	32.63220	3.369069	15.40280	16.62125	15.89463
4	-1620.630	44.37703*	3.389855	15.40741	17.00662	16.05293
5	-1604.134	29.16193	3.666410	15.48334	17.46332	16.28256
6	-1585.653	31.84675	3.900487	15.54155	17.90229	16.49446
7	-1573.090	21.08924	4.380908	15.65259	18.39409	16.75919
8	-1554.257	30.77076	4.660533	15.70765	18.82992	16.96795
9	-1536.932	27.53462	5.035006	15.77618	19.27921	17.19017
10	-1527.034	15.28936	5.825485	15.91102	19.79481	17.47870
11	-1509.772	25.89200	6.327175	15.98011	20.24467	17.70149
12	-1497.158	18.35801	7.183558	16.09070	20.73602	17.96578

* indicates lag order selected by the criterion

LR: sequential modified LR test statistic (each test at 5% level)

FPE: Final prediction error

AIC: Akaike information criterion

SC: Schwarz information criterion

HQ: Hannan-Quinn information criterion

Check of residuals (not shown) leads to VAR(4) for the levels

Cointegration Tests

Step 1 - Deterministic Components

Selected (0.05 level*) Number of Cointegrating Relations by Model

Data Trend:	None	None	Linear	Linear	Quadratic
Test Type	No Intercept	Intercept	Intercept	Intercept	Intercept
	No Trend	No Trend	No Trend	Trend	Trend
Trace	0	1	1	1	1
Max-Eig	1	1	1	0	1

*Critical values based on MacKinnon-Haug-Michelis (1999)

Information Criteria by Rank and Model

Data Trend:	None	None	Linear	Linear	Quadratic
Rank or No. of CEs	No Intercept	Intercept	Intercept	Intercept	Intercept
	No Trend	No Trend	No Trend	Trend	Trend
Log Likelihood by Rank (rows) and Model (columns)					
0	-1738.614	-1738.614	-1738.328	-1738.328	-1735.811
1	-1722.802	-1720.652	-1720.368	-1719.464	-1717.056
2	-1716.772	-1711.857	-1711.582	-1707.287	-1705.107
3	-1712.235	-1705.912	-1705.641	-1699.037	-1697.280
4	-1709.760	-1701.991	-1701.801	-1693.128	-1693.061
5	-1709.760	-1699.651	-1699.651	-1690.946	-1690.946
Akaike Information Criteria by Rank (rows) and Model (columns)					
0	15.91873	15.91873	15.95955	15.95955	15.98105
1	15.86842	15.85846*	15.89063	15.89147	15.90525
2	15.90279	15.87755	15.90114	15.88128	15.88837
3	15.95008	15.92132	15.93629	15.90509	15.90719
4	16.01524	15.98261	15.98962	15.94916	15.95723
5	16.10182	16.05758	16.05758	16.02550	16.02550
Schwarz Criteria by Rank (rows) and Model (columns)					
0	17.40896*	17.40896*	17.52429	17.52429	17.62029
1	17.50766	17.51261	17.60439	17.62013	17.69352
2	17.69106	17.69562	17.76392	17.77386	17.82566
3	17.88738	17.90332	17.94809	17.96160	17.99350
4	18.10155	18.12853	18.15045	18.16960	18.19257
5	18.33715	18.36743	18.36743	18.40986	18.40986

Step 2 - Tests

Sample (adjusted): 1987M06 2006M09

Included observations: 232 after adjustments

Trend assumption: No deterministic trend (restricted constant)

Series: AAUGVAL ABDGVAL AJPGVAL AUKGVAL AUSGVAL

Lags interval (in first differences): 1 to 3

Unrestricted Cointegration Rank Test (Trace)

Hypothesized No. of CE(s)	Eigenvalue	Trace Statistic	0.05 Critical Value	Prob.**
None *	0.155232	80.11174	76.97277	0.0282
At most 1	0.077637	40.97477	54.07904	0.4223
At most 2	0.049551	22.22535	35.19275	0.5803
At most 3	0.026358	10.43495	20.26184	0.5971
At most 4	0.018101	4.237829	9.164546	0.3779

Trace test indicates 1 cointegrating eqn(s) at the 0.05 level

* denotes rejection of the hypothesis at the 0.05 level

**MacKinnon-Haug-Michelis (1999) p-values

Unrestricted Cointegration Rank Test (Maximum Eigenvalue)

Hypothesized No. of CE(s)	Eigenvalue	Max-Eigen Statistic	0.05 Critical Value	Prob.**
None *	0.155232	39.13697	34.80587	0.0143
At most 1	0.077637	18.74943	28.58808	0.5126
At most 2	0.049551	11.79040	22.29962	0.6753
At most 3	0.026358	6.197116	15.89210	0.7656
At most 4	0.018101	4.237829	9.164546	0.3779

Max-eigenvalue test indicates 1 cointegrating eqn(s) at the 0.05 level

* denotes rejection of the hypothesis at the 0.05 level

**MacKinnon-Haug-Michelis (1999) p-values

Engle-Granger Two Step Estimator

1. Estimate the **cointegrating regression** and save the **residuals**,
($r = 1$ by design)

$$x_{1t} = \beta_0 + \beta_1 x_{2t} + \beta_2 x_{3t} + z_t$$

$$\hat{z}_t = x_{1t} - b_0 - b_1 x_{2t} - b_2 x_{3t}$$

2. Estimate the **VEC model** of the form (5) (equation by equation) using Least Squares; 3-VEC(2)

$$\begin{pmatrix} \Delta x_{1t} \\ \Delta x_{2t} \\ \Delta x_{3t} \end{pmatrix} = \begin{pmatrix} \pi_{10} \\ \pi_{20} \\ \pi_{30} \end{pmatrix} + \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} \hat{z}_{t-1} + \begin{pmatrix} \pi_{11} & \pi_{12}^{(1)} & \pi_{13}^{(1)} \\ \pi_{21}^{(1)} & \pi_{22}^{(1)} & \pi_{23}^{(1)} \\ \pi_{31}^{(1)} & \pi_{32}^{(1)} & \pi_{33}^{(1)} \end{pmatrix} \begin{pmatrix} \Delta y_{t-1} \\ \Delta x_{t-1} \\ \Delta w_{t-1} \end{pmatrix} \\ + \begin{pmatrix} \pi_{11}^{(2)} & \pi_{12}^{(2)} & \pi_{13}^{(2)} \\ \pi_{21}^{(2)} & \pi_{22}^{(2)} & \pi_{23}^{(2)} \\ \pi_{31}^{(2)} & \pi_{32}^{(2)} & \pi_{33}^{(2)} \end{pmatrix} \begin{pmatrix} \Delta y_{t-2} \\ \Delta x_{t-2} \\ \Delta w_{t-2} \end{pmatrix} + \begin{pmatrix} e_1 \\ e_2 \\ e_3 \end{pmatrix}$$

The estimator of π_1 and π_2 has a standard distribution as $\hat{z}_t$ is a stationary regressor

VEC Estimation by MLE

- ▶ A **one-step** estimation of (4)
- ▶ This is a **non-linear model**,
- ▶ The cointegrating vectors **must be identified**
 - ▶ Normalization
 - ▶ Further linear restrictions could be needed (zero restrictions or other forms)
- ▶ Estimates of the speed of adjustment parameters in α must be bounded between $[0,1)$.

VEC with $r=2$

- Example for two cointegrating vectors ($r = 2$)

$$\begin{pmatrix} \Delta x_{1t} \\ \Delta x_{2t} \\ \Delta x_{3t} \end{pmatrix} = \begin{pmatrix} \pi_{10} \\ \pi_{20} \\ \pi_{30} \end{pmatrix} + \begin{pmatrix} \alpha_{11} & \alpha_{12} \\ \alpha_{21} & \alpha_{22} \\ \alpha_{31} & \alpha_{32} \end{pmatrix} \begin{bmatrix} 1 & \beta_{21} & \beta_{31} \\ \beta_{12} & 1 & \beta_{32} \end{bmatrix} \begin{pmatrix} x_{1t-1} \\ x_{2t-1} \\ x_{3t-1} \end{pmatrix} \\ + \begin{pmatrix} \pi_{11}(1) & \pi_{12}(1) & \pi_{13}(1) \\ \pi_{21}(1) & \pi_{22}(1) & \pi_{23}(1) \\ \pi_{31}(1) & \pi_{32}(1) & \pi_{33}(1) \end{pmatrix} \begin{pmatrix} \Delta x_{1t-1} \\ \Delta x_{2t-1} \\ \Delta x_{3t-1} \end{pmatrix} \\ + \begin{pmatrix} \pi_{11}(2) & \pi_{12}(2) & \pi_{13}(2) \\ \pi_{21}(2) & \pi_{22}(2) & \pi_{23}(2) \\ \pi_{31}(2) & \pi_{32}(2) & \pi_{33}(2) \end{pmatrix} \begin{pmatrix} \Delta x_{1t-2} \\ \Delta x_{2t-2} \\ \Delta x_{3t-2} \end{pmatrix} + \begin{pmatrix} e_{1t} \\ e_{2t} \\ e_{3t} \end{pmatrix}$$

- First equation:

$$\begin{aligned} \Delta x_{1t} &= \pi_{10} + \alpha_{11}(x_{1t-1} + \beta_{21}x_{2t-1} + \beta_{31}x_{3t-1}) + \alpha_{12}(\beta_{12}x_{1t-1} + x_{2t-1} + \beta_{32}x_{3t-1}) \\ &\quad + \pi_{11}(1)\Delta x_{1t-1} + \pi_{12}(1)\Delta x_{2t-1} + \pi_{13}(1)\Delta x_{3t-1} \\ &\quad + \pi_{11}(2)\Delta x_{1t-2} + \pi_{12}(2)\Delta x_{2t-2} + \pi_{13}(2)\Delta x_{3t-2} + e_{1t} \end{aligned}$$

VEC with $r=1$

- ▶ Example for one cointegrating vectors ($r = 1$)

$$\begin{pmatrix} \Delta x_{1t} \\ \Delta x_{2t} \\ \Delta x_{3t} \end{pmatrix} = \begin{pmatrix} \pi_{10} \\ \pi_{20} \\ \pi_{30} \end{pmatrix} + \begin{pmatrix} \alpha_{11} \\ \alpha_{21} \\ \alpha_{31} \end{pmatrix} \begin{bmatrix} 1 & \beta_{21} & \beta_{31} \end{bmatrix} \begin{pmatrix} x_{1t-1} \\ x_{2t-1} \\ x_{3t-1} \end{pmatrix} \\ + \begin{pmatrix} \pi_{11}(1) & \pi_{12}(1) & \pi_{13}(1) \\ \pi_{21}(1) & \pi_{22}(1) & \pi_{23}(1) \\ \pi_{31}(1) & \pi_{32}(1) & \pi_{33}(1) \end{pmatrix} \begin{pmatrix} \Delta x_{1t-1} \\ \Delta x_{2t-1} \\ \Delta x_{3t-1} \end{pmatrix} \\ + \begin{pmatrix} \pi_{11}(2) & \pi_{12}(2) & \pi_{13}(2) \\ \pi_{21}(2) & \pi_{22}(2) & \pi_{23}(2) \\ \pi_{31}(2) & \pi_{32}(2) & \pi_{33}(2) \end{pmatrix} \begin{pmatrix} \Delta x_{1t-2} \\ \Delta x_{2t-2} \\ \Delta x_{3t-2} \end{pmatrix} + \begin{pmatrix} e_{1t} \\ e_{2t} \\ e_{3t} \end{pmatrix}$$

- ▶ First equation:

$$\begin{aligned} \Delta x_{1t} &= \pi_{10} + \alpha_{11}(x_{1t-1} + \beta_{21}x_{2t-1} + \beta_{31}x_{3t-1}) \\ &\quad + \pi_{11}(1)\Delta x_{1t-1} + \pi_{12}(1)\Delta x_{2t-1} + \pi_{13}(1)\Delta x_{3t-1} \\ &\quad + \pi_{11}(2)\Delta x_{1t-2} + \pi_{12}(2)\Delta x_{2t-2} + \pi_{13}(2)\Delta x_{3t-2} + e_{1t} \end{aligned}$$

Estimation of VEC Model - Long-run estimates only shown

Sample (adjusted): 1987M06 2006M09

Included observations: 232 after adjustments

Standard errors in () & t-statistics in []

Cointegrating Eq:	CointEq1					
AAUGVAL(-1)	1.000000					
ABDGVAL(-1)	1.644154 (0.37565) [4.37685]					
AJPGVAL(-1)	-1.392016 (0.32956) [-4.22384]					
AUKGVAL(-1)	-0.322184 (0.26321) [-1.22407]					
AUSGVAL(-1)	-0.223884 (0.29111) [-0.76906]					
C	-78.32383 (22.9928) [-3.40645]					
Error Correction:	D(AAUGVAL)	D(ABDGVAL)	D(AJPGVAL)	D(AUKGVAL)	D(AUSGVAL)	
CointEq1	-0.045849 (0.01983) [-2.31168]	0.000989 (0.01138) [0.08694]	0.042049 (0.01369) [3.07219]	0.017361 (0.01989) [0.87270]	-0.046588 (0.01641) [-2.83828]	

